Integration 10X Hive

2024-08-05 15:06:42 +0200

Loading package

```
suppressMessages(library(formatR))
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(ggplot2))
suppressMessages(library(scales))
Sys.setenv(LANG = "en")
```

```
knitr::opts_chunk$set(tidy.opts = list(width.cutoff = 60), tidy = TRUE)
```

Loading data

```
data10X <- readRDS("../Data/lavage_cells10xV2.rds")

data10X$Annotation <- data10X$seurat_clusters
levels(data10X$Annotation) <- c("10X C1 - AM", "10X C10 - LT",
    "10X C8 - LT", "10X C4 - MoMac", "10X C11 - LT", "10X C5 - Neu",
    "10X C2 - AM", "10X C7 - LT", "10X C9 - LT", "10X C13 - Epith",
    "10X C6 - Neu", "10X C12 - Mast", "10X C3 - AM")

data10X$Annotation <- factor(data10X$Annotation, levels = c("10X C1 - AM",
    "10X C2 - AM", "10X C3 - AM", "10X C4 - MoMac", "10X C5 - Neu",
    "10X C6 - Neu", "10X C7 - LT", "10X C8 - LT", "10X C9 - LT",
    "10X C10 - LT", "10X C11 - LT", "10X C12 - Mast", "10X C13 - Epith"))


dataHive <- readRDS("../Data/Lavage_cellsHive.rds")
dataHive$Annotation <- dataHive$seurat_clusters
levels(dataHive$Annotation) <- c("Hive C1 - AM", "Hive C7 - LT",
    "Hive C3 - AM", "Hive C8 - LT", "Hive C6 - Neu", "Hive C4 - MoMac",
    "Hive C10 - Other", "Hive C2 - AM", "Hive C5 - Neu", "Hive C9 - Eos",
    "Hive C11 - Mast")
dataHive$Annotation <- factor(dataHive$Annotation, levels = c("Hive C1 - AM",
    "Hive C2 - AM", "Hive C3 - AM", "Hive C4 - MoMac", "Hive C5 - Neu",
    "Hive C6 - Neu", "Hive C7 - LT", "Hive C8 - LT", "Hive C9 - Eos",
    "Hive C10 - Other", "Hive C11 - Mast"))
```

listing metadata

```
Metadata_10x <- data10X@meta.data[, c(1, 6, 8)]

Metadata_Hive <- dataHive@meta.data[, c("orig.ident", "Annotation",
    "seurat_clusters")]

metadata.list <- list(Metadata_10x, Metadata_Hive)
```

listing count

```
Count_10x <- data10X@assays$RNA@counts

Count_Hive <- dataHive@assays$RNA@counts

data.list <- list(Count_10x, Count_Hive)

Integrate_object <- CreateSeuratObject(counts = data.list, meta.data = metadata.list)
```

```
Integrate_object <- merge(data10X, dataHive)
```

```
Integrate_object$Group <- Integrate_object$orig.ident
Integrate_object$Group <- as.factor(Integrate_object$Group)
levels(Integrate_object$Group) <- c("Hive", "Hive", "Hive", "Hive",
    "10X", "10X", "10X", "10X")
```

Integration

```
Integrate_object[["RNA"]] <- split(Integrate_object[["RNA"]],
    f = Integrate_object$Group)
```

```
Integrate_object <- NormalizeData(Integrate_object)
Integrate_object <- FindVariableFeatures(Integrate_object)
Integrate_object <- ScaleData(Integrate_object)
Integrate_object <- RunPCA(Integrate_object)
```

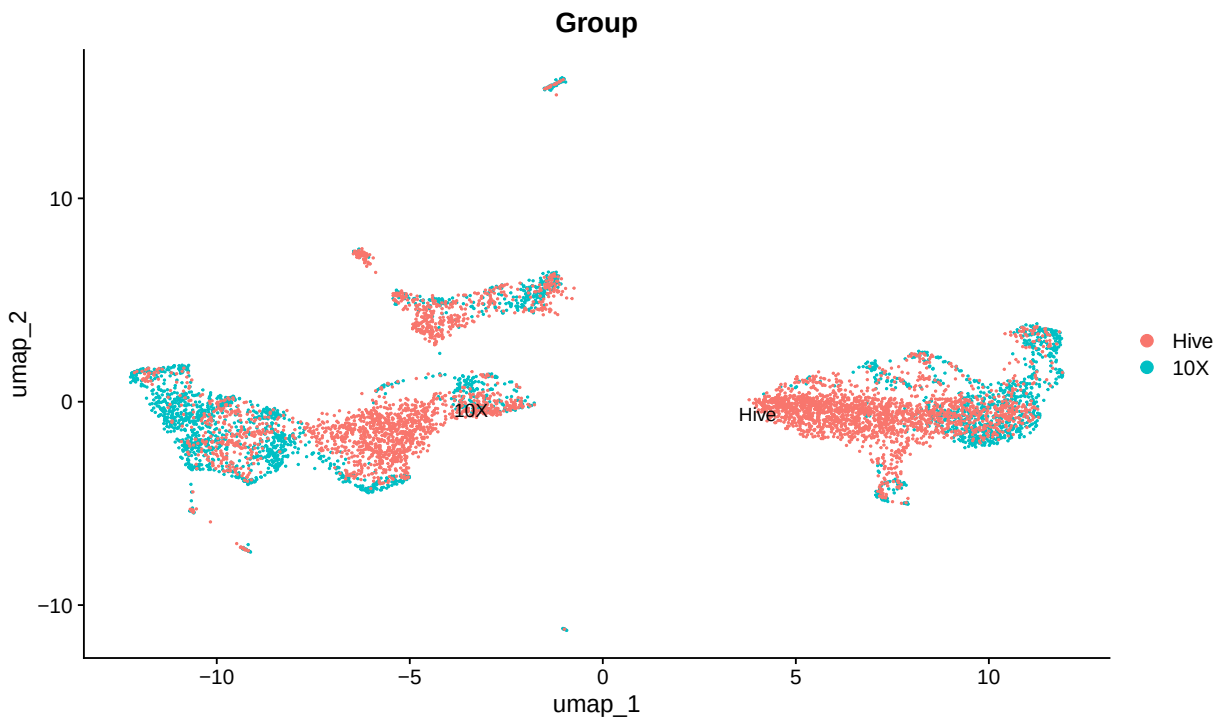
RPCA

```
Integrate_object <- IntegrateLayers(Integrate_object, method = RPCAIntegration,
    orig.reduction = "pca", new.reduction = "integrated.rpca")

Integrate_object <- FindNeighbors(Integrate_object, reduction = "integrated.rpca",
    dims = 1:30)
```

```
Integrate_object <- FindClusters(Integrate_object, resolution = 1)
Integrate_object <- RunUMAP(Integrate_object, dims = 1:30, reduction = "integrated.rpca")
```

```
DimPlot(Integrate_object, reduction = "umap", group.by = c("Group"),
    label = T)
```



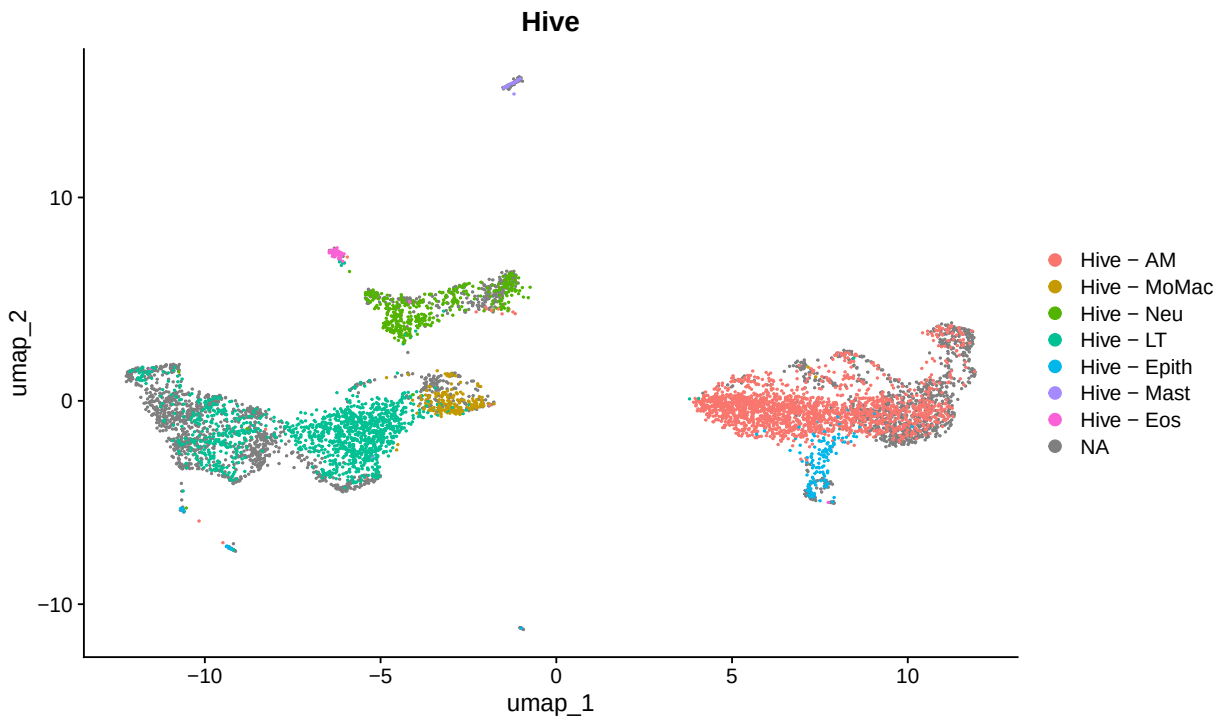
Merging Annotation

```
Integrate_object$Annotation_merged <- Integrate_object$Annotation
Integrate_object$Annotation_merged <- as.factor(Integrate_object$Annotation_merged)
levels(Integrate_object$Annotation_merged) <- c("10X - AM", "10X - LT",
    "10X - LT", "10X - Mast", "10X - Epith", "10X - AM", "10X - AM",
    "10X - MoMac", "10X - Neu", "10X - Neu", "10X - LT", "10X - LT",
    "10X - LT", "Hive - AM", "Hive - Epith", "Hive - Mast", "Hive - AM",
    "Hive - AM", "Hive - MoMac", "Hive - Neu", "Hive - Neu",
    "Hive - LT", "Hive - LT", "Hive - Eos")
Integrate_object$Annotation_merged <- factor(Integrate_object$Annotation_merged,
    levels = c("10X - AM", "10X - MoMac", "10X - Neu", "10X - LT",
        "10X - Epith", "10X - Mast", "Hive - AM", "Hive - MoMac",
        "Hive - Neu", "Hive - LT", "Hive - Epith", "Hive - Mast",
        "Hive - Eos"))
```

Umap Hive

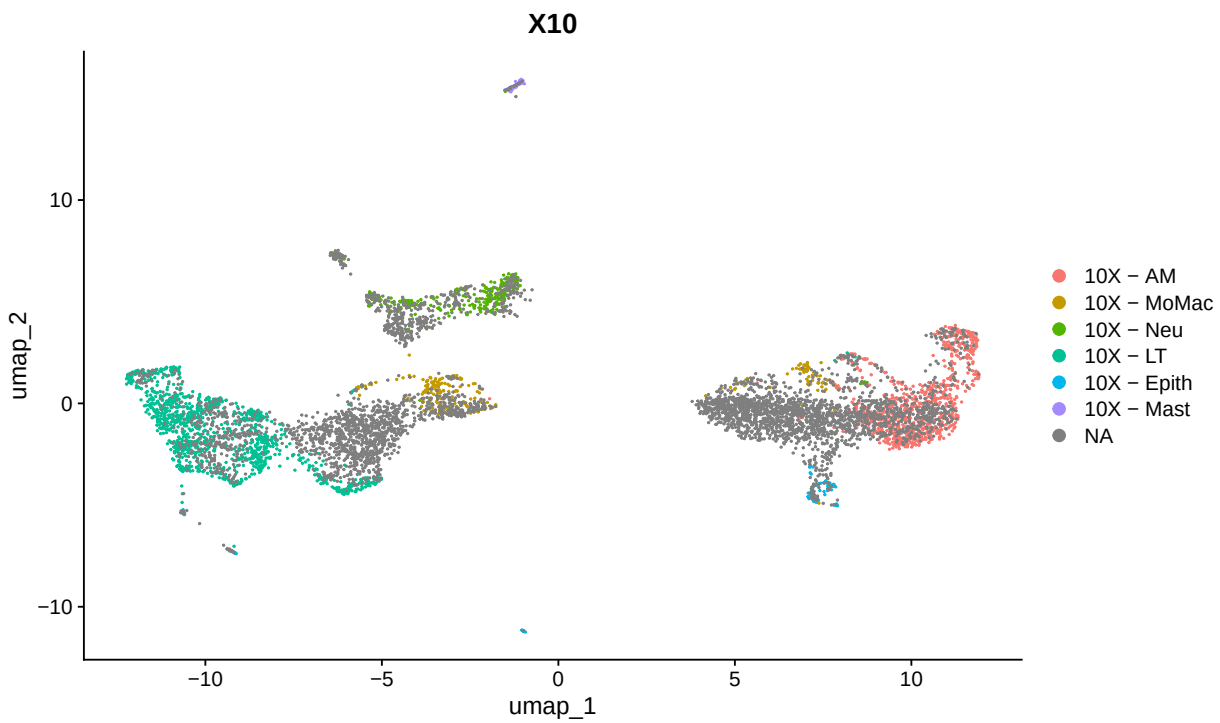
```
`%notin%` <- function(x, y) !(x %in% y)

Integrate_object$Hive <- Integrate_object$Annotation_merged
Integrate_object$Hive[Integrate_object$Hive %notin% c("Hive - AM",
    "Hive - MoMac", "Hive - Neu", "Hive - LT", "Hive - Epith",
    "Hive - Mast", "Hive - Eos")] <- NA
DimPlot(Integrate_object, reduction = "umap", group.by = c("Hive"))
```




```
# ggsave('plot_clusters_pdf/integration_Hive.pdf', width =
# 9, height = 6)
```

```
Integrate_object$X10 <- Integrate_object$Annotation_merged
Integrate_object$X10[Integrate_object$X10 %notin% c("10X - AM",
    "10X - MoMac", "10X - Neu", "10X - LT", "10X - Epith", "10X - Mast")] <- NA
DimPlot(Integrate_object, reduction = "umap", group.by = c("X10"),
    cols = hue_pal()(7))
```




```
# ggsave('plot_clusters_pdf/integration_10X.pdf', width =
# 9, height = 6)
```

```
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 10 x64 (build 19045)
##
## Matrix products: default
##
##
## locale:
## [1] LC_COLLATE=French_Belgium.utf8  LC_CTYPE=French_Belgium.utf8
## [3] LC_MONETARY=French_Belgium.utf8 LC_NUMERIC=C
## [5] LC_TIME=French_Belgium.utf8
##
## time zone: Europe/Brussels
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] scales_1.3.0       ggplot2_3.5.1      Seurat_5.1.0       SeuratObject_5.0.2
## [5] sp_2.1-4           dplyr_1.1.4        formatR_1.14
##
## loaded via a namespace (and not attached):
##   [1] deldir_2.0-4           pbapply_1.7-2          gridExtra_2.3
```

```
##   [4] rlang_1.1.4            magrittr_2.0.3         RcppAnnoy_0.0.22
##   [7] spatstat.geom_3.2-9    matrixStats_1.3.0      ggridges_0.5.6
##  [10] compiler_4.4.1         png_0.1-8              vctrs_0.6.5
##  [13] reshape2_1.4.4         stringr_1.5.1          pkgconfig_2.0.3
##  [16] fastmap_1.2.0          labeling_0.4.3         utf8_1.2.4
##  [19] promises_1.3.0         rmarkdown_2.27         tinytex_0.51
##  [22] purrr_1.0.2            xfun_0.45              jsonlite_1.8.8
##  [25] goftest_1.2-3          highr_0.11             later_1.3.2
##  [28] spatstat.utils_3.0-5   irlba_2.3.5.1          parallel_4.4.1
##  [31] cluster_2.1.6          R6_2.5.1               ica_1.0-3
##  [34] spatstat.data_3.1-2    stringi_1.8.4          RColorBrewer_1.1-3
##  [37] reticulate_1.38.0      parallelly_1.37.1      lmtest_0.9-40
##  [40] scattermore_1.2        Rcpp_1.0.12            knitr_1.47
##  [43] tensor_1.5             future.apply_1.11.2    zoo_1.8-12
##  [46] sctransform_0.4.1      httpuv_1.6.15          Matrix_1.7-0
##  [49] splines_4.4.1          igraph_2.0.3           tidyselect_1.2.1
##  [52] abind_1.4-5            rstudioapi_0.16.0      yaml_2.3.8
##  [55] spatstat.random_3.2-3  codetools_0.2-20       miniUI_0.1.1.1
##  [58] spatstat.explore_3.2-7 listenv_0.9.1          lattice_0.22-6
##  [61] tibble_3.2.1           plyr_1.8.9             withr_3.0.0
##  [64] shiny_1.8.1.1          ROCR_1.0-11            evaluate_0.24.0
##  [67] Rtsne_0.17             future_1.33.2          fastDummies_1.7.3
##  [70] survival_3.6-4         polyclip_1.10-6        fitdistrplus_1.1-11
##  [73] pillar_1.9.0           KernSmooth_2.23-24     plotly_4.10.4
##  [76] generics_0.1.3         RcppHNSW_0.6.0         munsell_0.5.1
##  [79] globals_0.16.3         xtable_1.8-4           glue_1.7.0
##  [82] lazyeval_0.2.2         tools_4.4.1            data.table_1.15.4
##  [85] RSpectra_0.16-1        RANN_2.6.1             leiden_0.4.3.1
##  [88] dotCall64_1.1-1        cowplot_1.1.3          grid_4.4.1
##  [91] tidyr_1.3.1            colorspace_2.1-0       nlme_3.1-164
##  [94] patchwork_1.2.0        cli_3.6.3              spatstat.sparse_3.1-0
##  [97] spam_2.10-0            fansi_1.0.6            viridisLite_0.4.2
## [100] uwot_0.2.2             gtable_0.3.5           digest_0.6.36
## [103] progressr_0.14.0       ggrepel_0.9.5          farver_2.1.2
## [106] htmlwidgets_1.6.4      htmltools_0.5.8.1      lifecycle_1.0.4
## [109] httr_1.4.7             mime_0.12              MASS_7.3-60.2
```